

Pre-Trained Inference

Module 7 Week B — When Not to Fine-Tune

Get to know Lana — Fun Fact

AI.SPIRE

AI TINKERERS

**The global community for
hands-on AI builders.**

Curated meetups, hackathons, dinners, and demos for people shipping with foundation models. Bring working code, share what broke, and leave with patterns you can ship.



Recommended for you: Seattle Durango Denver Lehi

Apply to attend

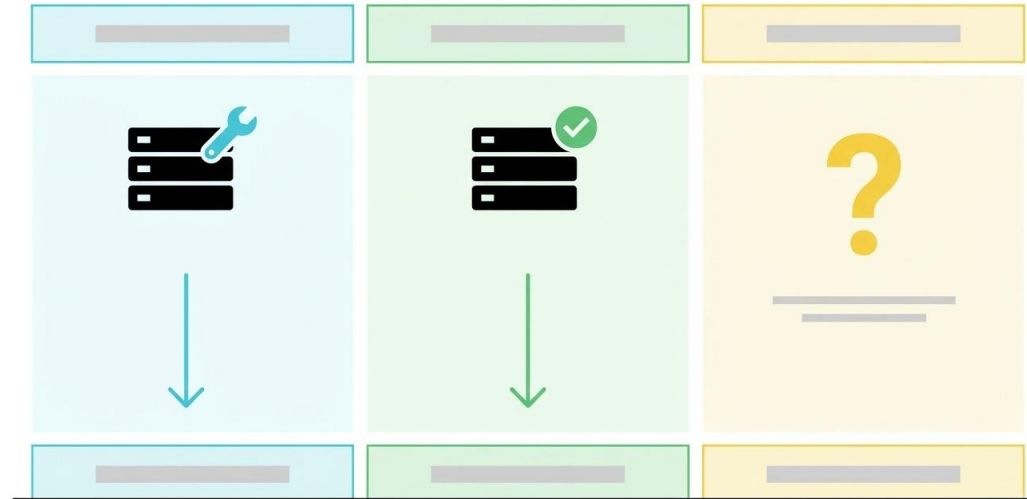
Start a city

Sponsor the room

I'm an AI Tinkerer

- Joined in 2024
- Demo'ed 3 times
- **I highly recommend starting an Amman AIT chapter!!**

Pre-training to Inference



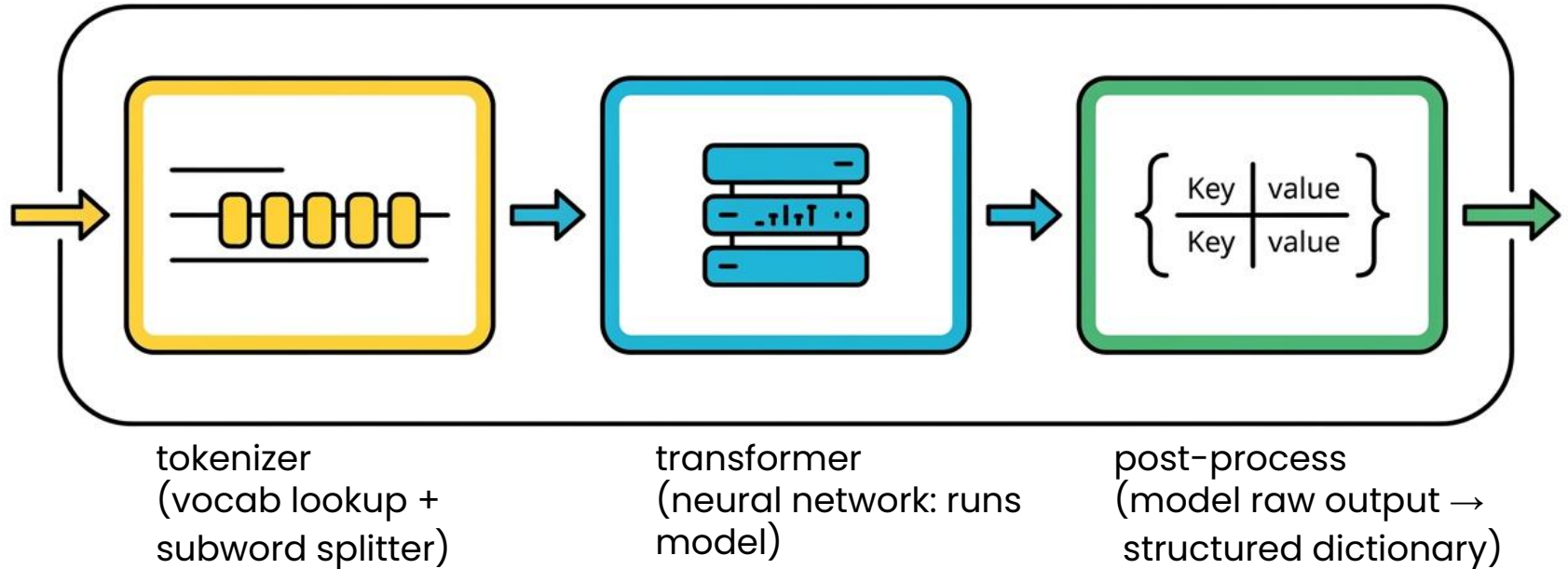
Fine-tuning

Pre-trained
inference

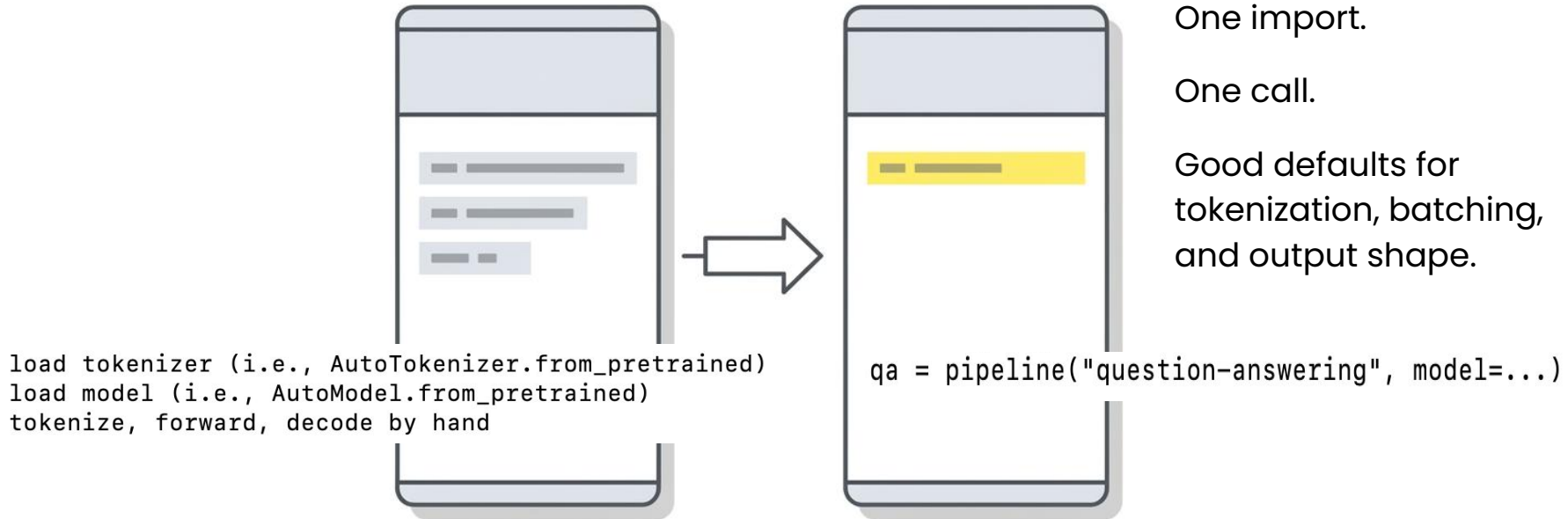
The decision:
which one do you
use for your task?

Pipeline Abstraction

Hugging Face `pipeline()` object



What The Pipeline Does



Three Lines For Question Answering

```
from transformers import pipeline

qa = pipeline("question-answering")

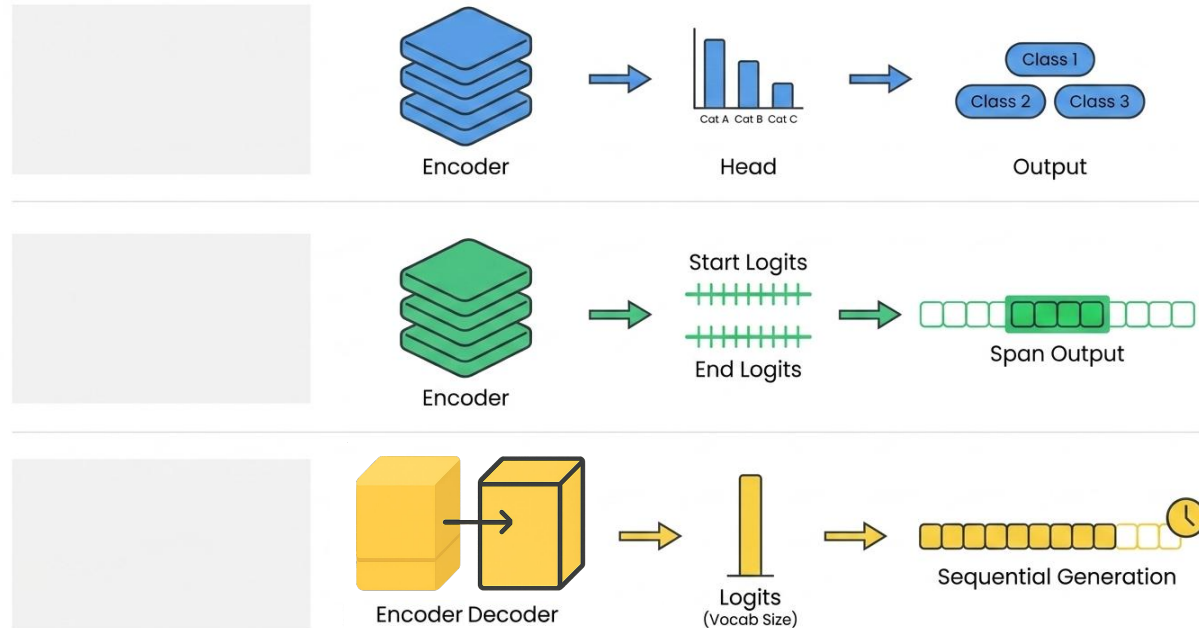
result = qa(question="Who founded Petra Telecom?",
             context="Petra Telecom was founded in Amman in 2003 by Layla Haddad.")
```

distilbert-base-cased-distilled-squad

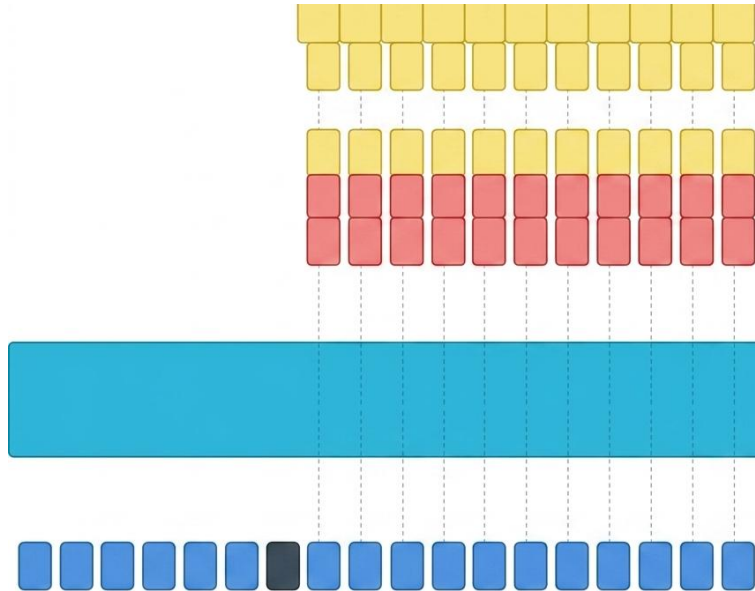
↓ print(result)

```
{
  answer: "Layla Haddad",
  score: 0.94,
  start: 47,
  end: 59
}
```

Three Task Heads

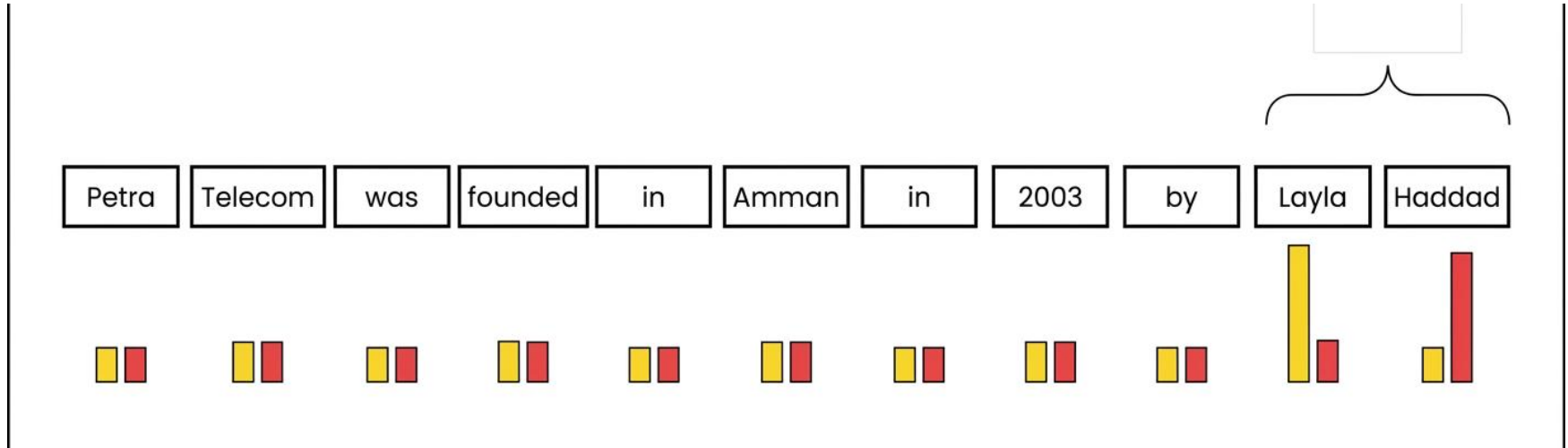


Span Prediction



Two heads on top of the encoder: one for start, one for end.

Picking The Span



Cannot Hallucinate



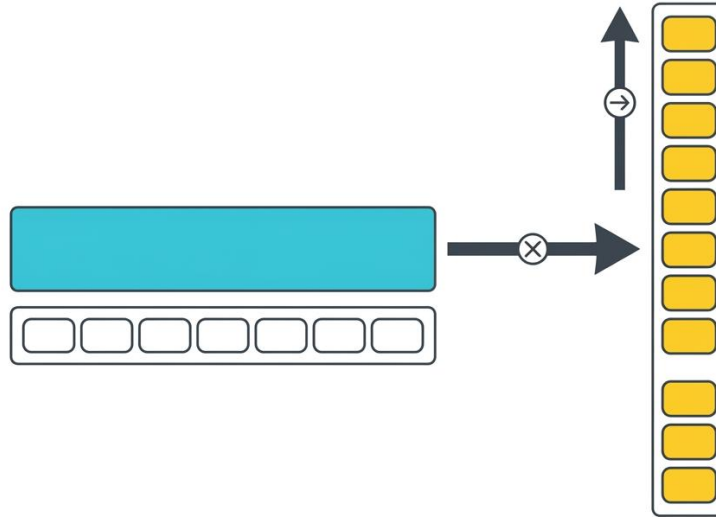
No hallucination also means:

no synthesis,

no rephrasing,

no answer from multiple passages.

Encoder-Decoder

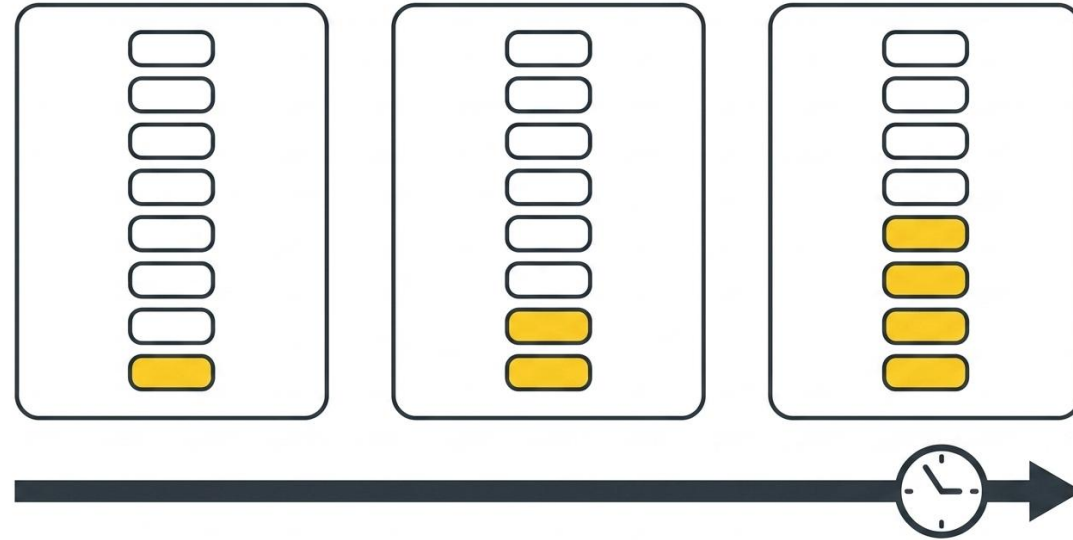


Encoder reads.

Decoder writes (generates).

Cross-attention is the bridge between the input and the output.

Decoder: One Token At A Time



Generation Decoding: Deterministic or Stochastic

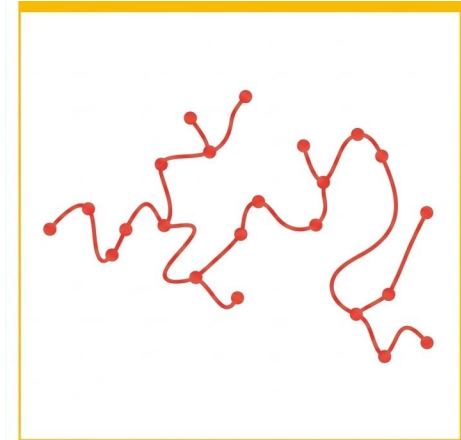
Deterministic

Beam search is repeatable and stable.

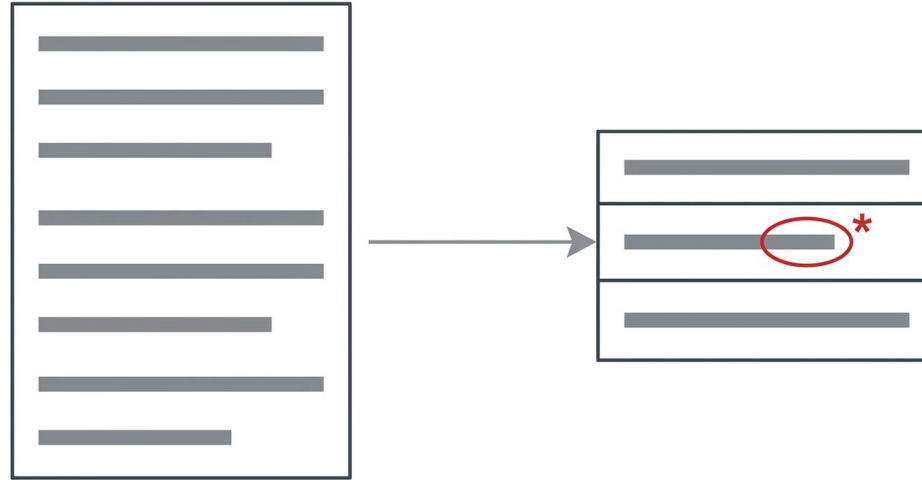


Stochastic (non-deterministic)

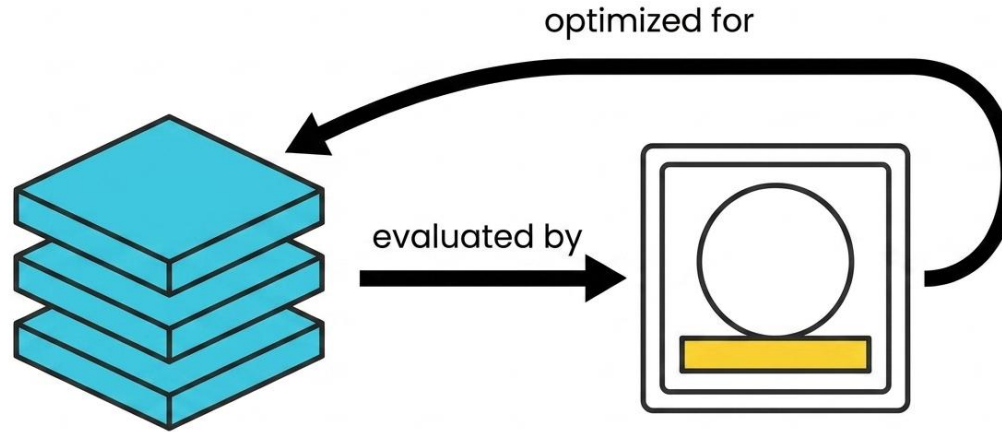
Sampling uses randomness and gives output variation.



Generation Can Hallucinate



Metric Is A Values Statement

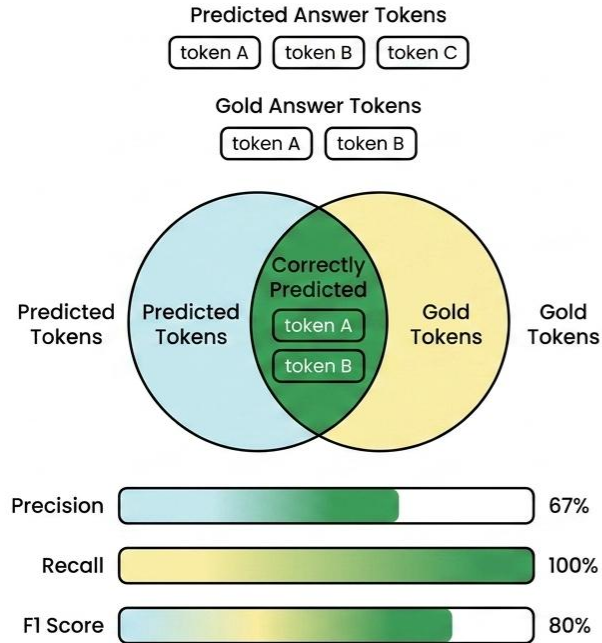


Exact-Match

One if normalized predicted
equals normalized gold.
Otherwise zero. No partial credit.

predicted	gold	result
Layla Haddad	Layla Haddad	✓
layla haddad	Layla Haddad	✓
Mrs. Layla Haddad	Layla Haddad	✗

Token-F1



Bag-of-tokens overlap.

Harmonic mean of precision and recall.

Partial credit.

Answer Normalization



Summarization: ROUGE



ROUGE-1

"did you use the right vocabulary"

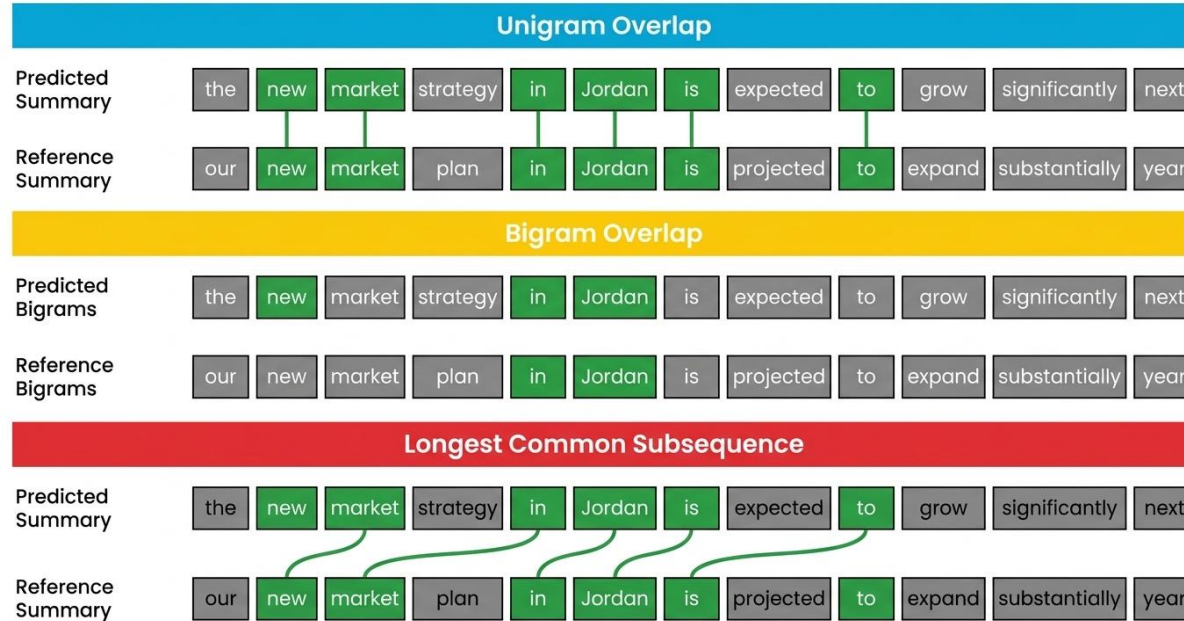
ROUGE-2

"did you put words together the same way"

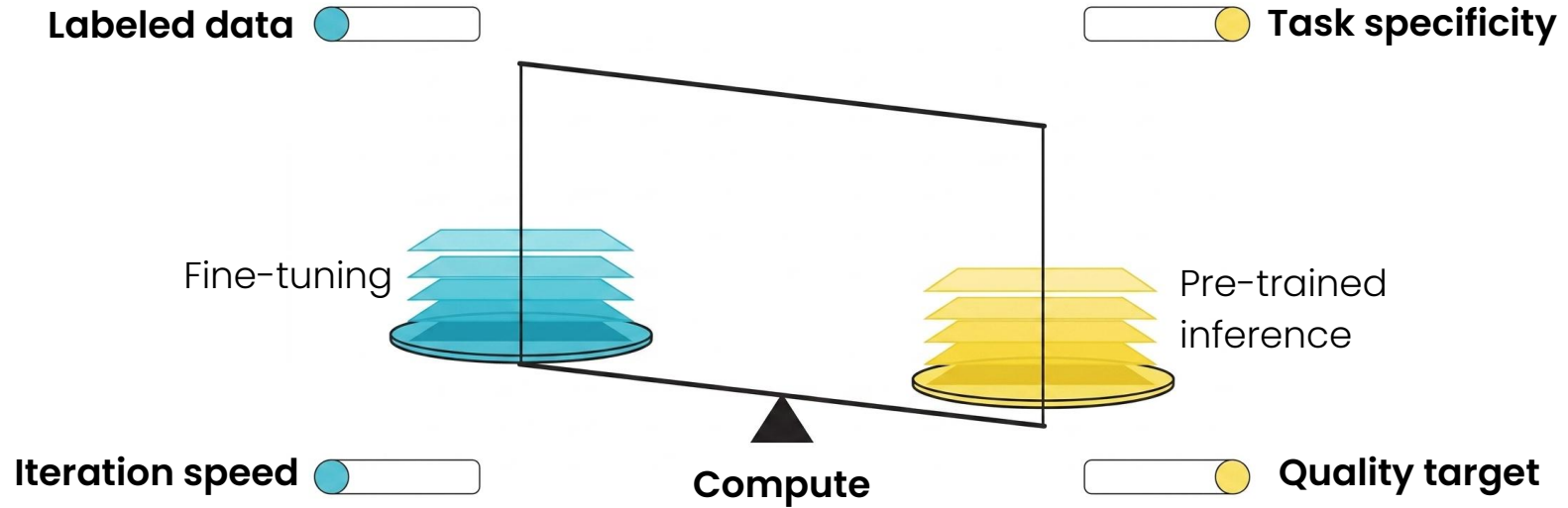
ROUGE-L

"did you keep the same overall flow"

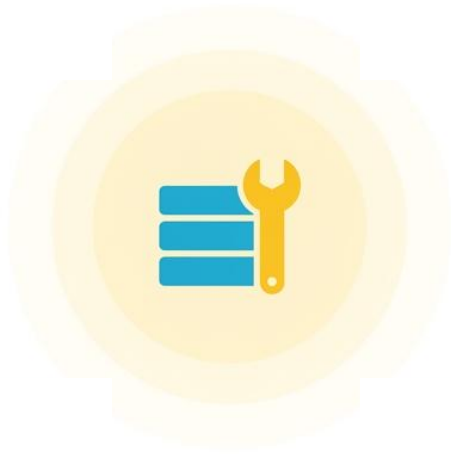
ROUGE



The Five-Factor Decision



When Fine-Tuning Works



Niche task.

Labels available.

Quality gap is big.

Long-lived service.

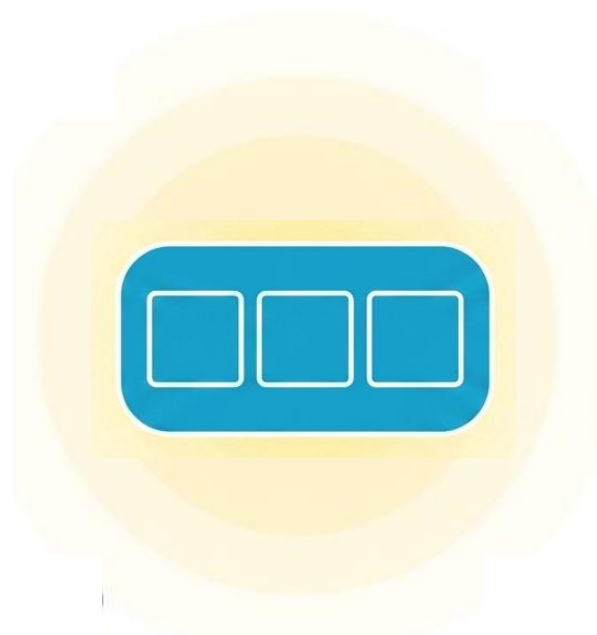
When Pre-Trained Wins

Generic task.

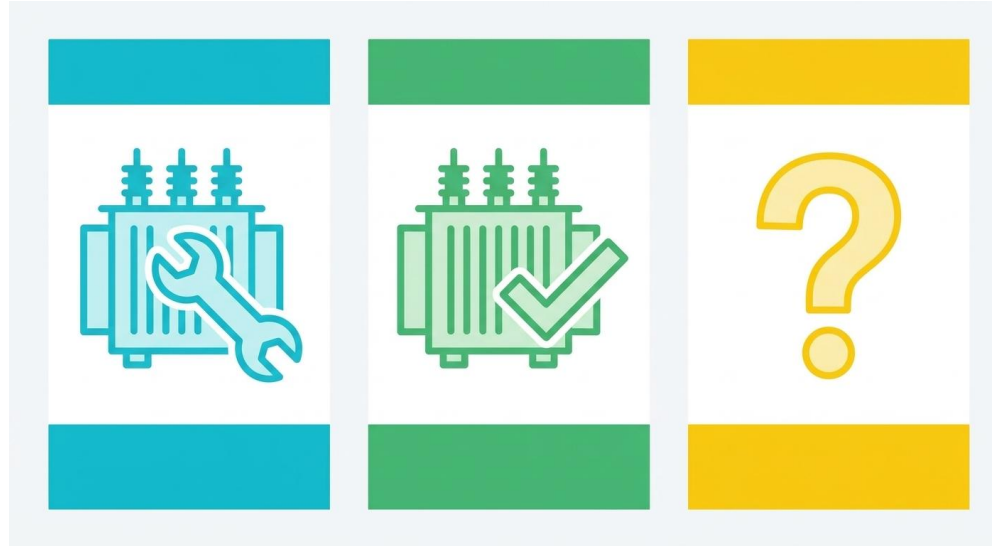
No labels.

Iteration speed matters.

Acceptable quality.



This Week



Questions?